

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Solution Requirements

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Med/Low)
1	Authentication	Secure login via ServiceNow native authentication engine, supporting Multi-Factor Authentication (MFA) and single sign-on (SSO) credential validation.	High
2	Authorization levels	Role-based granular authorization: bb1 (Read ACL), bb2 (Create ACL), bb3 (Write ACL), bb4 (Delete ACL), with administrative overrides for 'admin' and 'security_admin'.	High
3	External interfaces	Native integration with ServiceNow Service Portal, REST Table APIs, and application navigator lists without security bypass.	Medium
4	Transactions processing	Dynamic evaluation of server-side Script-Controlled ACLs on custom table 'u_institution_details' matching Branch choice value ('EEE') during CRUD transactions.	High
5	Reporting	Restricted list views and audit reports ensuring branch users only see their own department's data; admins can export cross-branch compliance reports.	Medium
6	Business rules, etc.	Pre-record evaluation conditions (Branch == 'EEE') combined with server-side script logic (gs.hasRole) executing in isolated scope prior to granting access.	High
7	Compliance to laws or regulations	Adherence to FERPA (academic data privacy), institutional data governance standards, and least-privilege security frameworks.	High
8	Other	Automated system logging of unauthorized access attempts and elevated privilege tracking within ServiceNow System Logs.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ACL rule script evaluation overhead must remain imperceptible during batch record fetching and table list	Record list query execution overhead < 200ms; page load

		loading.	< 2.0s
2	Scalability	Security architecture must maintain linear response times as the institution details record table scales across departments.	Supports 50,000+ student records and 500+ concurrent role-based queries
3	Security & Data Privacy	Zero unauthorized cross-department data exposure; defense-in-depth through server-side ACL script enforcement.	100% record denial for unauthorized queries; zero data leakage
4	Reliability & Availability	Access Control Lists must continuously govern record operations without runtime JavaScript exceptions or unhandled failures.	99.9% platform availability; zero ACL syntax/script runtime errors
5	Usability & Accessibility	Clean list view rendering indicating 'No records to display' for restricted records; accessibility compliant UI.	Complies with WCAG 2.1 AA; clear visual indication of permitted actions
6	Other (Maintainability)	Modular scripting and standardized naming conventions for easy replication across new academic branches (e.g., MECH, CIVIL).	Code maintainability index > 85; configuration turnaround < 1 hr per department